Project 2: Modelling daily cycling demand in Edinburgh

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1 Introduction

This report analyses a rich longitudinal dataset of approximately 2,180 daily observations spanning from 2020 to 2025, captured by a network of automatic cycling counters positioned at strategic locations across the City of Edinburgh. Our core modelling goal is to develop, rigorously evaluate, and compare a sequence of increasingly complex linear regression models to accurately predict daily cycling demand using temporal and meteorological covariates. By quantifying how cycling volumes vary with weather, seasonality, day of the week, and long-term drift, the resulting optimal model will be used to answer specific planning questions of practical interest. Ultimately, understanding these demand dynamics is invaluable for Edinburgh's transport planners, providing the data-driven insights required to evaluate reliable seasonal usage, assess temperature sensitivity, and optimise long-term infrastructure investment and resource allocation over the next decade.

2 Exploratory Data Analysis

The dataset contains 2180 daily observations from 1 January 2020 to 31 December 2025. Three preparatory transformations were applied. First, `month` was converted to an ordered factor with levels January through December for correct plot ordering. Second, `dow` was re-specified with explicit levels (Sun, Mon, …, Sat) to fix Sunday as the reference level. Third, a numeric `trend` variable was derived as integer days elapsed since 2020-01-01, giving us a simple linear proxy for long-run drift.

Table 1: Summary statistics for daily count and temperature variables.

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max
count	2180	10815.47	4728.88	1264.00	7476.00	10531.00	13720.00	43864
temp_mean	2180	9.69	4.79	-8.08	6.25	9.82	13.54	23
temp_min	2180	5.92	4.98	-14.00	2.00	6.00	10.00	18
temp_max	2180	13.15	5.31	-3.00	9.00	13.00	17.00	31

Daily counts range from 1,264 to 43,864 with a mean of 1.0815×10^4 (SD = 4729). The mean exceeds the median (1.0531×10^4), confirming right skew likely driven by exceptional summer weekdays. Mean temperature averages 9.7°C. The Pearson correlation between mean temperature and count is 0.21.

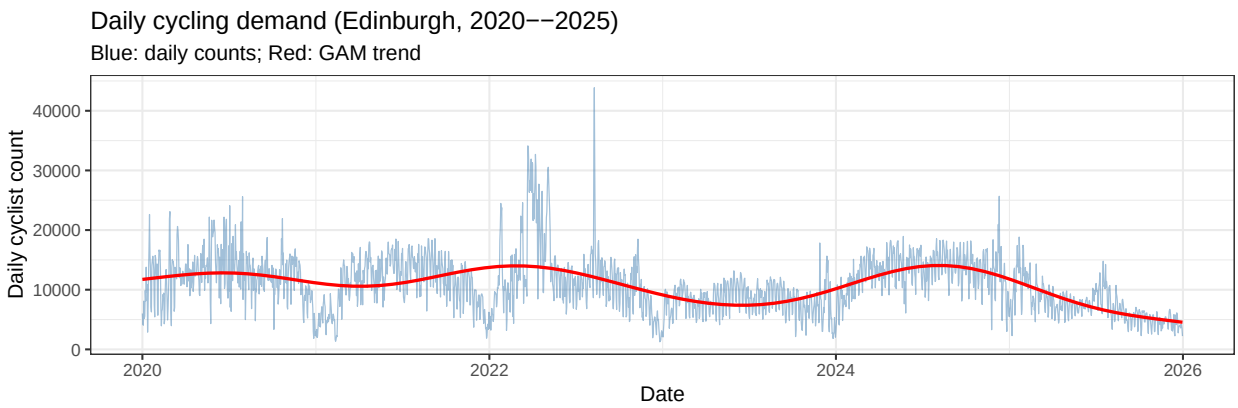

Figure 1: Daily cycling demand over 2020–2025. The GAM smoother highlights the strong annual cycle, with peaks each summer and troughs in winter.

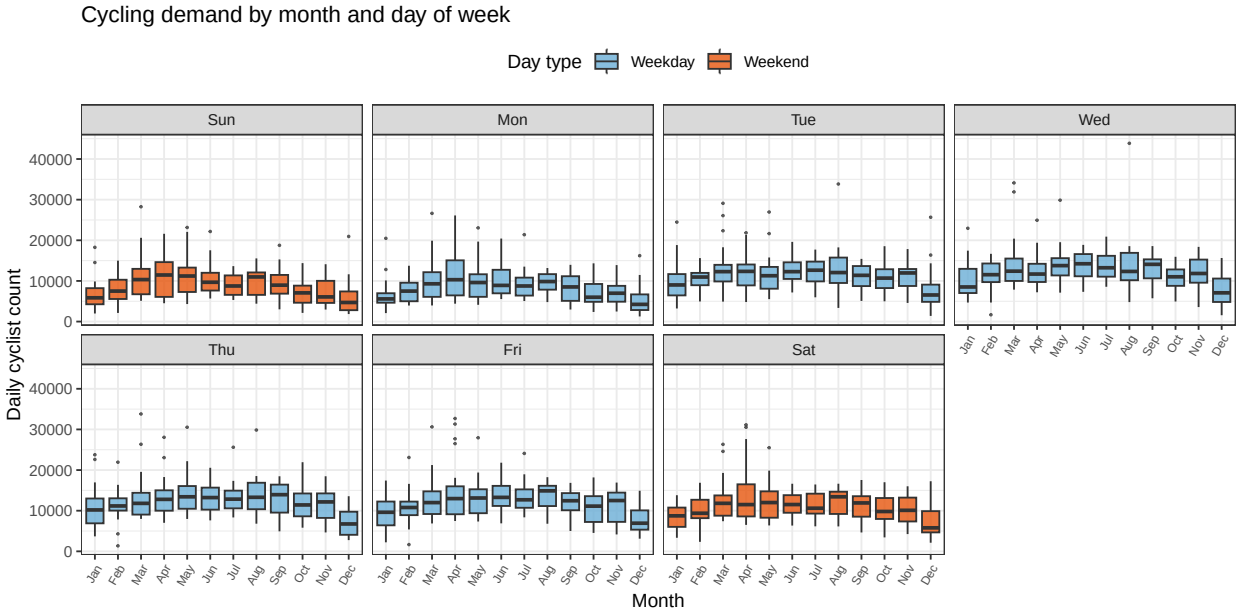

Figure 2: Cycling demand by month and day of week. Weekend medians (orange) exceed weekday medians (blue) most noticeably in summer, suggesting a seasonal leisure premium that a simple additive day-of-week factor won't fully capture.

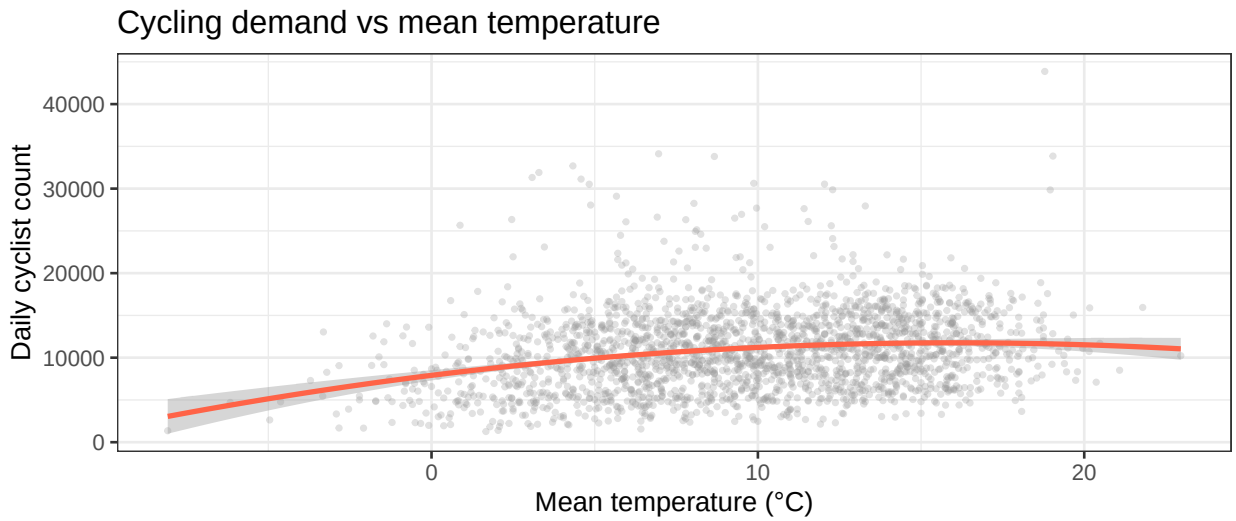

Figure 3: Count versus mean temperature with quadratic smoother. The concave shape motivates a quadratic temperature term in M2.

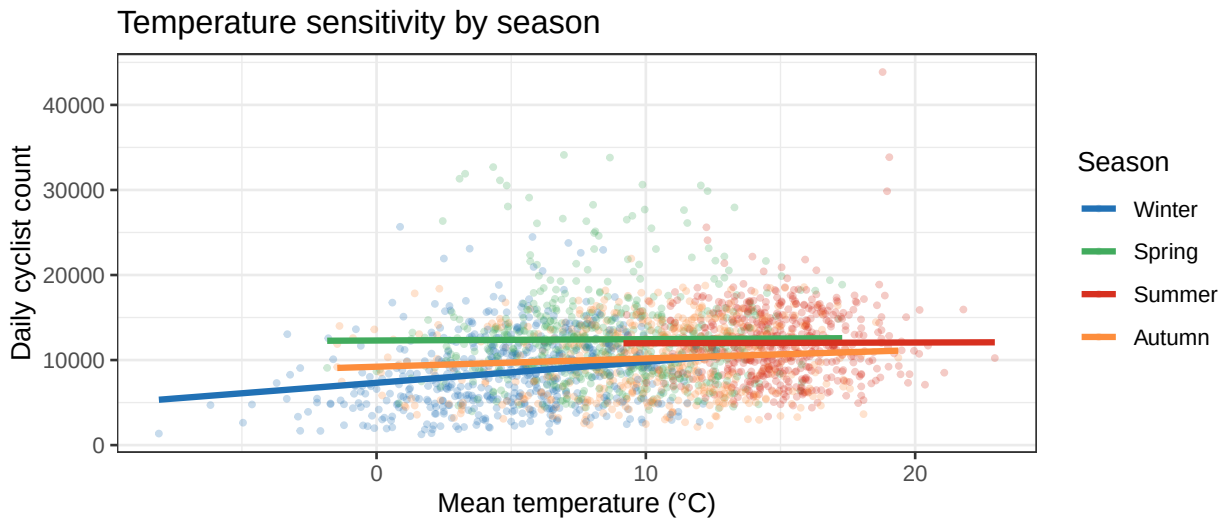

Figure 4: Temperature sensitivity by meteorological season. Winter and autumn slopes are steepest, while the nearly flat summer slope suggests demand saturation at higher temperatures.

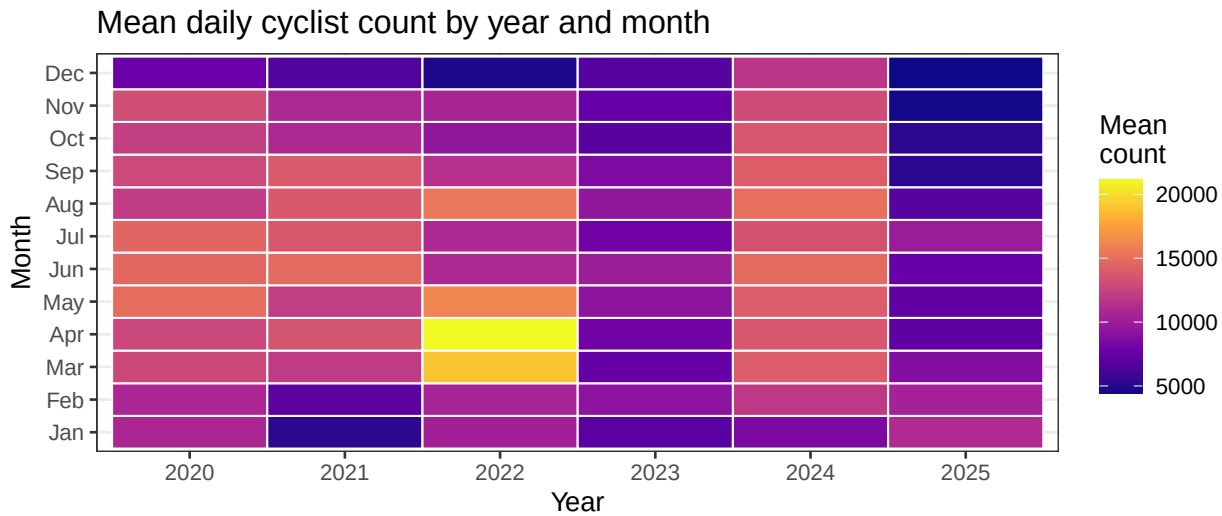

Figure 5: Mean daily count by year and month. Counts in 2020 and 2021 do not appear notably depressed relative to surrounding years, consistent with Figure 1, but a gradual downward drift in demand is visible from 2022 onwards.

Figures 1–5 highlight four key patterns. The time series shows a clear annual cycle with summer peaks and winter troughs. The faceted boxplots show that the gap between weekend and weekday medians is largest in summer, suggesting a seasonal interaction that an additive day-of-week factor will miss. The temperature scatter reveals a concave relationship: count rises steeply from around 0°C to roughly 15°C and then flattens, motivating a quadratic term. Season-stratified slopes confirm that temperature sensitivity is highest in winter and autumn, where each extra degree has the biggest impact on demand; the nearly flat summer slope suggests cyclists are already out regardless of temperature. The heatmap does not show a clear demand dip during 2020–2021, consistent with Figure 1, but reveals a gradual downward drift in demand post-2022, motivating both a linear trend and a COVID indicator to separate any pandemic regime shift from the secular trend.

3 Baseline Model M0

3.1 Mathematical Specification

M0 treats month as a continuous numeric predictor and uses a binary weekend indicator:

$$\text{count}_i = \beta_0 + \beta_1\,\text{temp_mean}_i + \beta_2\,\text{weekend}_i + \beta_3\,\text{month_num}_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2) \tag{1}$$

The numeric encoding forces a monotone linear seasonal ramp, which is a deliberate oversimplification for a baseline benchmark.

3.2 Coefficients

M0 coefficient estimates. All predictors highly significant ($p < 0.001$).

Term	Estimate	Std.Error	t.value	p.value
(Intercept)	10385.819	267.015	38.90	0
temp_mean	247.179	20.769	11.90	0

weekend	-1246.888	214.172	-5.82	0
month_num	-247.184	28.810	-8.58	0

All three predictors are significant ($p < 0.001$). Temperature has a positive coefficient (+247 cyclists per °C). The weekend coefficient is negative (−1247), which makes sense since commuter cycling dominates in Edinburgh. The negative month coefficient forces a monotone decrease from January to December, which clearly contradicts the summer peaks we saw in the EDA. M0 achieves $R^2 = 0.089$ (from `summary(m0)$r.squared`), explaining only 8.9% of daily count variance.

3.3 Diagnostics

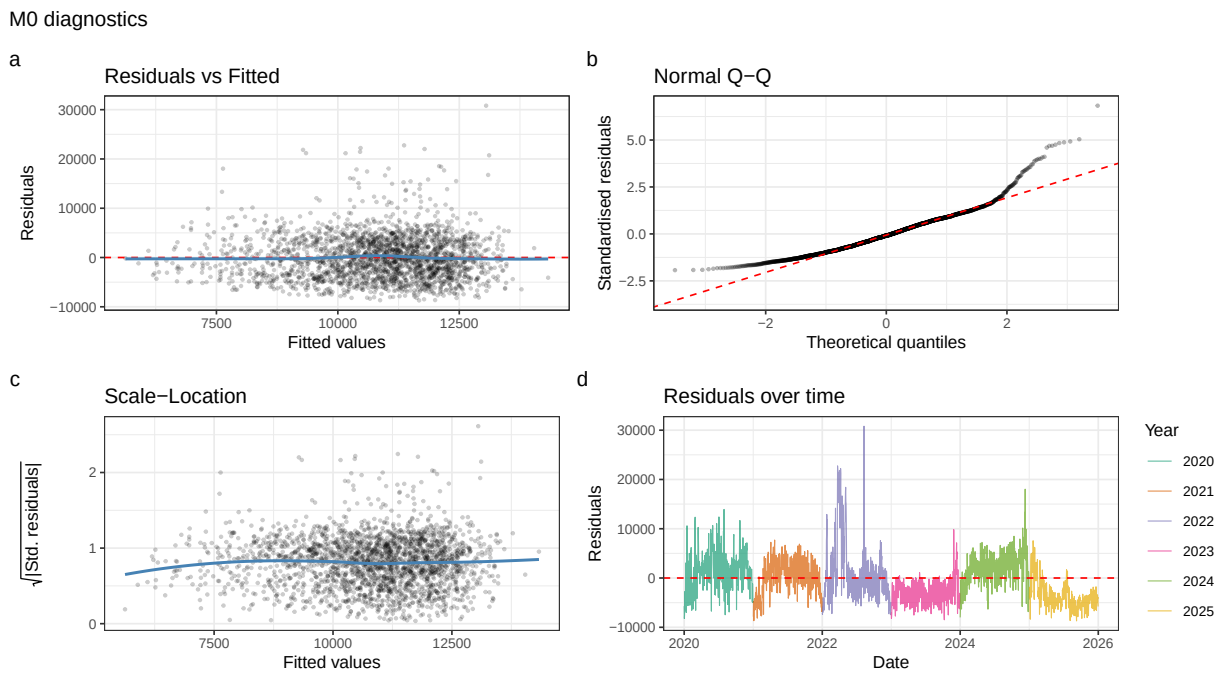

Figure 6: M0 diagnostics. (a) Curved residuals-vs-fitted indicating nonlinearity. (b) Heavy right tail in QQ. (c) Funnel-shaped heteroscedasticity. (d) Residuals over time show no strong systematic temporal structure.

The curved pattern in (a), the heavy QQ tail in (b), and the funnel shape in (c) all point to model misspecification. Together, they motivate M1: switching month and day-of-week to factors for flexible seasonal and weekly profiles, plus adding a linear trend.

4 Proposed Models M1–M3

4.1 Model M1: Trend and Factor Encodings

$$\text{count}_i = \beta_0 + \beta_1\,\text{temp_mean}_i + \sum_{m=2}^{12} \gamma_m\,\mathbf{1}[\text{month}_i = m] + \beta_2\,\text{trend}_i + \sum_{d=2}^{7} \lambda_d\,\mathbf{1}[\text{dow}_i = d] + \varepsilon_i \tag{2}$$

where January and Sunday are reference levels. M1 replaces numeric month with a factor so each month gets its own intercept, adds a full day-of-week factor (which subsumes the binary weekend variable, since

including both would cause perfect collinearity), and introduces a linear trend for secular drift. M1 achieves $R^2 = 0.282$, a big jump from 0.089. Factor month captures the asymmetric seasonal curve; factor day-of-week distinguishes each day rather than just weekday vs weekend; the trend absorbs long-run drift.

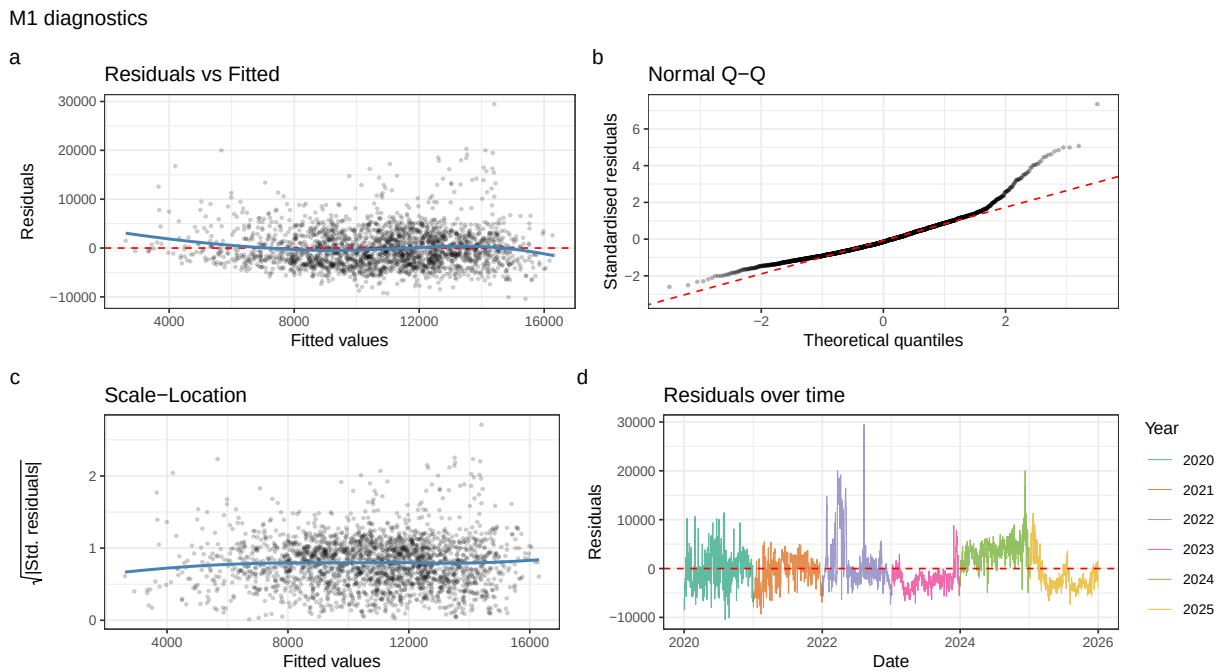

Figure 7: M1 diagnostics. (a) Reduced curvature. (b) Right tail persists. (c) Funnel-shaped heteroscedasticity remains. (d) Residuals over time show a positive band in 2022 and negative bands in 2023 and 2025.

4.2 Model M2: Nonlinear Temperature

The EDA scatter and M1 diagnostics both suggest a concave temperature-count relationship. M2 replaces the linear temperature term with a quadratic. We write temp_mean_i as T_i for brevity:

$$\text{count}_i = \beta_0 + \beta_1 T_i + \beta_2 T_i^2 + \sum_{m=2}^{12} \gamma_m \, \mathbf{1}[\text{month}_i = m] + \beta_3 \, \text{trend}_i + \sum_{d=2}^{7} \lambda_d \, \mathbf{1}[\text{dow}_i = d] + \varepsilon_i \tag{3}$$

where $T_i \equiv \text{temp_mean}_i$. If $\beta_2 < 0$, the relationship is concave (demand rises then flattens). M2 achieves $R^2 = 0.285$.

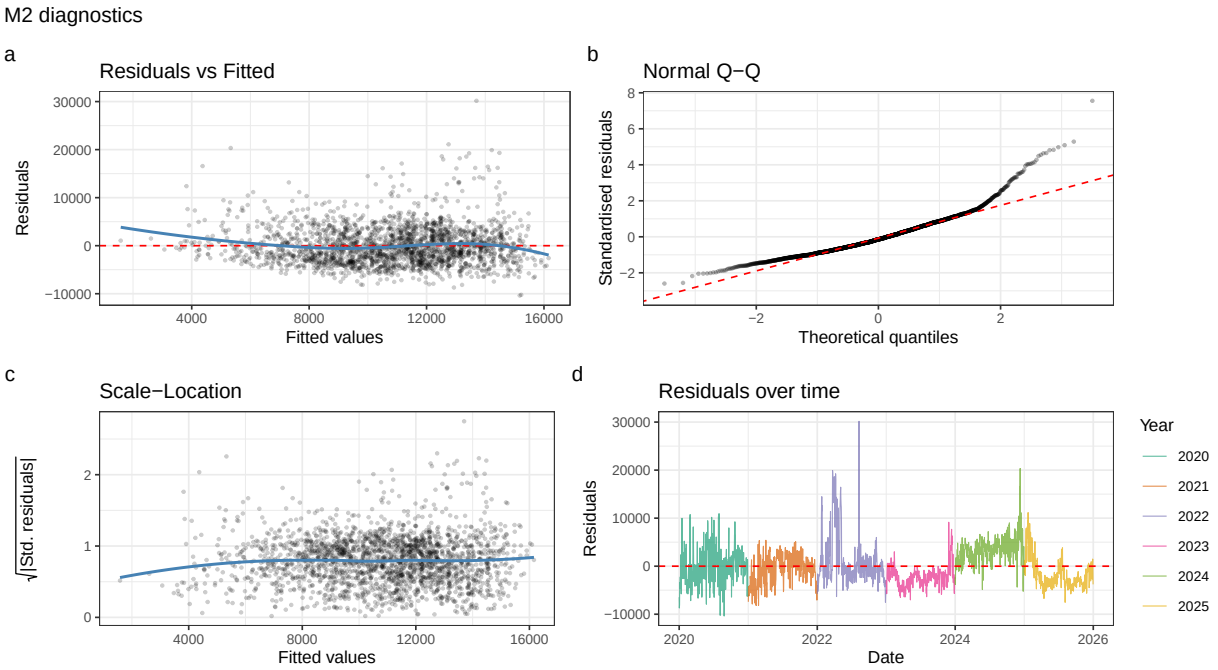

Figure 8: M2 diagnostics. (a) Flatter residuals. (b) Right tail persists. (c) Funnel-shaped heteroscedasticity: variance increases with fitted values. (d) Residuals over time show a positive band in 2022 and negative bands in 2023 and 2025, motivating a further correction in M3.

The most persistent diagnostic signal is the structured pattern in residuals over time (Figure 8d): a positive band in 2022 suggests M2 under-predicts during the post-COVID recovery, while negative bands in 2023 and 2025 indicate over-prediction in later years.

4.3 Model M3: COVID-19 Indicator

Diagnostic justification. The residuals-over-time plot for M2 (Figure 8d) shows a structured temporal pattern: a positive band in the first half of 2022 suggests under-prediction during the post-COVID recovery spike, while negative bands in 2023, 2024, and 2025 indicate systematic over-prediction in later years. The residuals for 2020–2021 themselves are relatively well-behaved; the problem is that the trend coefficient is being distorted by the pandemic period, pulling the estimated slope too steeply negative and causing the model to over-predict post-2022 demand. We considered three candidate extensions. A *log-transformation* would help with the Q-Q right tail but complicates back-transformation and makes coefficients harder to interpret for planning (we test this empirically in Section 6). A *temperature*×season interaction would pick up the differential slopes in Figure 4, but this doesn't address the temporal mean shift associated with the COVID period. A *COVID indicator* directly accounts for the pandemic regime change and helps separate the pandemic mean level from the secular trend coefficient, which matters for the planning questions. Before fitting M3, we expected the indicator to: (i) shift the trend coefficient by decoupling the pandemic-period mean level from the secular drift; and (ii) thereby attenuate the structured residual banding in 2023–2025 and reduce the under-prediction spike in the first half of 2022.

M3 adds a binary indicator $\text{is_covid}_i = \mathbf{1}[\text{year}_i \in \{2020, 2021\}]$:

$$\text{count}_i = \beta_0 + \beta_1 T_i + \beta_2 T_i^2 + \sum_{m=2}^{12} \gamma_m \, \mathbf{1}[\text{month}_i = m] + \beta_3 \, \text{trend}_i + \sum_{d=2}^{7} \lambda_d \, \mathbf{1}[\text{dow}_i = d] + \phi \, \text{is_covid}_i + \varepsilon_i \quad (4)$$

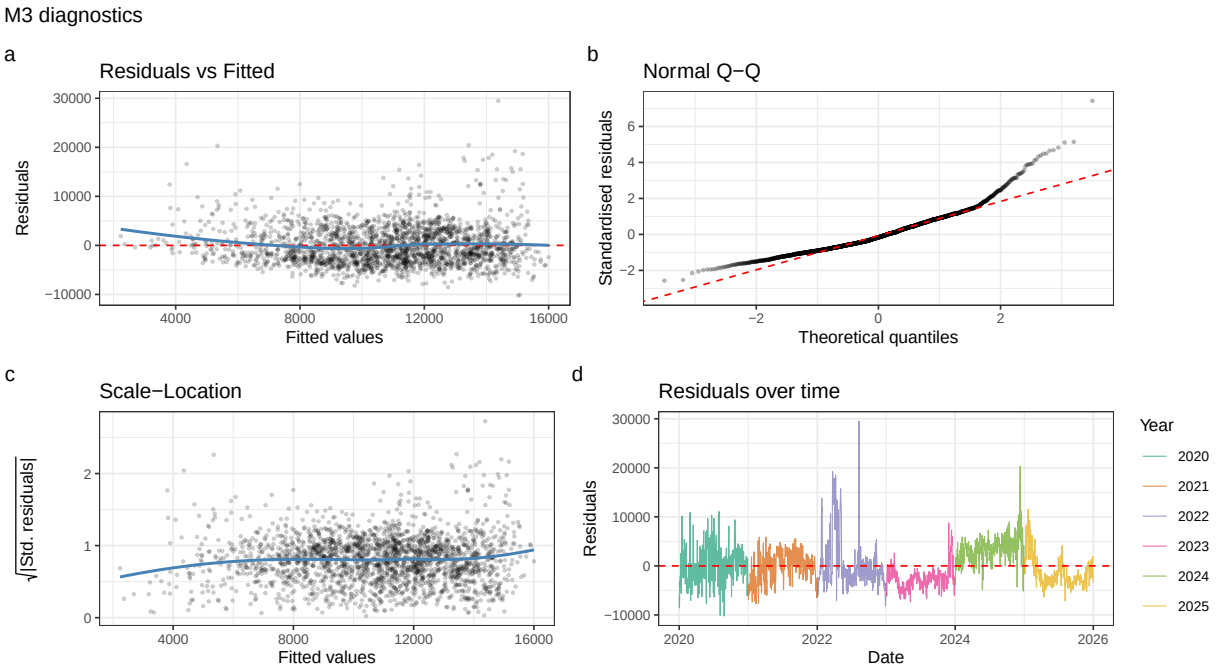

Figure 9: M3 diagnostics. (a) Most homoscedastic residuals-vs-fitted of the four models. (b) Heavy right tail improves marginally. (c) Funnel shape persists but improves marginally. (d) No major improvement visible; the 2022 positive band and 2023–2025 negative bands persist.

Figure 9d shows no major improvement in the temporal residual structure (R^2: 0.285 → 0.293). The 2022 positive band and the negative bands in 2023–2025 are largely unchanged. Right-tail skew and heteroscedasticity also remain. Changing the predictors won't fix these; we'd need a different response model like a log-transform or GLM (see Section 6).

4.4 Final Model Coefficients

M3 coefficient estimates. January and Sunday are reference levels.

Term	Estimate	Std.Error	t.value	p.value
(Intercept)	8909.038	526.251	16.93	0.00e+00
temp_mean	346.207	64.612	5.36	1.00e-07
temp_mean2	-11.794	3.518	-3.35	8.15e-04
month_fFeb	971.743	426.565	2.28	2.28e-02
month_fMar	3238.710	422.569	7.66	0.00e+00
month_fApr	3362.474	437.312	7.69	0.00e+00
month_fMay	2649.472	469.937	5.64	0.00e+00
month_fJun	2452.664	521.210	4.71	2.70e-06
month_fJul	2096.028	557.691	3.76	1.76e-04
month_fAug	2598.481	552.130	4.71	2.70e-06
month_fSep	1587.852	502.588	3.16	1.60e-03
month_fOct	600.749	464.139	1.29	1.96e-01
month_fNov	1443.418	435.456	3.31	9.32e-04
month_fDec	-1015.898	423.980	-2.40	1.67e-02
trend	-2.860	0.246	-11.64	0.00e+00
dowMon	-444.858	320.436	-1.39	1.65e-01
dowTue	2324.454	320.538	7.25	0.00e+00

dowWed	3253.853	320.249	10.16	0.00e+00
dowThu	3228.941	320.390	10.08	0.00e+00
dowFri	3007.835	320.539	9.38	0.00e+00
dowSat	2001.878	320.170	6.25	0.00e+00
is_covid	-1575.049	323.994	-4.86	1.30e-06

All coefficients have the expected signs. Monthly intercepts peak in spring (March to April) and are lowest in December. Wednesday to Friday show the highest demand relative to Sunday, while Monday is notably lower. The `is_covid` coefficient is -1575 ($p < 0.001$), meaning demand was about 1575 cyclists/day lower on average during 2020–2021 after controlling for everything else.

5 Evaluation

5.1 Statistical Performance

Table 1: Leave-one-year-out CV scores for M0–M3. Lower is better.

Model	RMSE	MAE	DS	IS
M0	4740.9	3805.6	18.05	25630.5
M1	4604.5	3669.2	18.07	25856.4
M2	4585.5	3654.1	18.06	25647.4
M3	4764.7	3851.9	18.15	25827.8

What each score measures. RMSE penalises large errors quadratically, making it sensitive to outliers. MAE treats all errors equally and gives a more robust central-tendency measure. DS is a proper scoring rule that rewards *calibrated uncertainty*: a model with good point predictions but overconfident intervals will score poorly. IS evaluates 95% prediction intervals directly, penalising both excessive width and coverage failures. Empirical PI coverage is roughly correct across models: M0 = 96%, M1 = 93.8%, M2 = 93.7%, M3 = 93.7% (nominal 95%).

Metric agreement. The metrics do not all agree. RMSE and MAE both select M2 as best, reflecting its improved point predictions from the quadratic temperature term. DS and IS, however, both select M0 as best; these proper scoring rules penalise miscalibrated prediction intervals, and the added complexity of M1–M3 widens intervals without a proportionate gain in coverage accuracy. M3 performs worst overall, which is mostly a problem with how the CV works here: the LOYO approach assumes the folds are roughly exchangeable, but the COVID indicator means pandemic and non-pandemic folds behave very differently. With only two COVID years (2020 and 2021), when either is held out, the `is_covid` coefficient has to be estimated from just one remaining COVID year, which increases variance.

Model selection: M3. Despite M2's better RMSE/MAE and M0's better DS/IS, we select M3 as the final model because the main goal isn't just predictive accuracy but answering planning questions. Without the COVID indicator, the trend coefficient mixes pandemic stagnation with secular drift, giving a misleading estimate. The CV disadvantage comes from the LOYO scheme not handling the 2020–2021 break well, not from M3 being a genuinely worse model. That said, if we only cared about prediction accuracy on future non-pandemic data, M2 would be the better pick on point-prediction metrics.

Table 2: CV RMSE and DS by month for the final model (M3).

Month	RMSE	DS
Jan	4782.8	18.01
Feb	4019.2	17.57

Mar	5974.0	19.14
Apr	6409.0	19.51
May	5074.4	18.32
Jun	4006.0	17.57
Jul	3723.9	17.44
Aug	5264.7	18.45
Sep	4434.3	17.79
Oct	4418.0	17.79
Nov	4577.0	17.91
Dec	5012.7	18.18

5.2 Practical Performance

5.2.1 Temperature Effects and Nonlinearity

CV comparison of M3 variants using different quadratic temperature predictors.

Temperature variable	RMSE	DS	IS
temp_mean	4764.7	18.15	25827.8
temp_min	4811.4	18.15	25812.7
temp_max	4739.8	18.14	25580.1

Despite `temp_max` achieving slightly lower RMSE, the differences are small (under 80 cyclists) and **`temp_mean` is retained** for two reasons. First, cycling trips in Edinburgh are spread across the day (morning commute, lunchtime, evening return), so the daily mean better represents the temperature conditions cyclists actually experience than the daily maximum, which peaks briefly around midday. Second, `temp_mean` and `temp_max` are highly correlated (Pearson $r = 0.95$), so the CV differences are within noise. Using `temp_mean` also keeps the model more comparable with transport studies that typically report results per degree of mean daily temperature.

For the quadratic model, $d\hat{y}/dT = \hat{\beta}_1 + 2\hat{\beta}_2 T$ where $T \equiv$ temp_mean. With $\hat{\beta}_1 = 346.21$ and $\hat{\beta}_2 = -11.794$:

- **At 5°C**, a 1°C increase is associated with approximately **228 additional cyclists per day**.
- **At 15°C**, a 1°C increase is associated with approximately **-8 additional cyclists per day**.

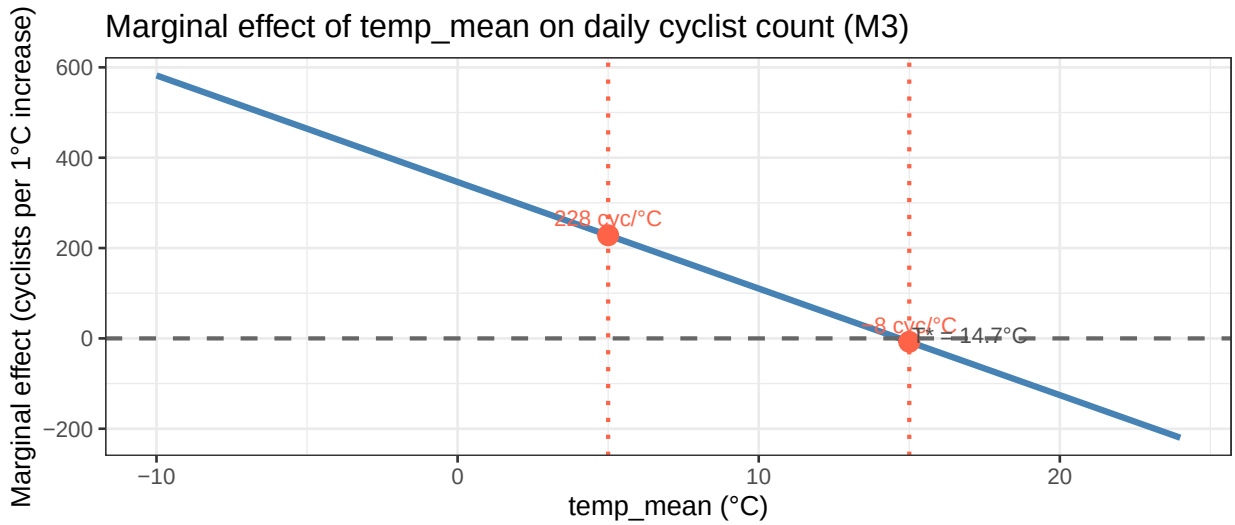

Figure 10: Marginal effect of mean temperature on daily cyclist count (M3). The effect diminishes to zero at $T^* \approx 14.7$°C.

The marginal effect reaches zero at $T^* = 14.7$°C. Below this threshold, infrastructure that makes cycling more comfortable in cold conditions (e.g. covered parking, all-weather paths) would give the biggest ridership return. M3 doesn't include a temperature-by-season interaction, so the 228 cyclists/°C figure at 5°C is a year-round average. Looking at Figure 4, winter slopes are noticeably steeper, so the true winter-specific sensitivity is probably higher than this pooled estimate. The IQR in July (5094 cyclists/day) versus January (5343 cyclists/day) also confirms higher variability in summer. A `temp_mean:is_summer` interaction could capture this; we'd expect a higher temperature coefficient in spring when cyclists start resuming habitual journeys, compared with winter when only the most committed cyclists are out. For planners, this means cold-weather infrastructure investments could deliver larger returns than the pooled estimate implies.

5.2.2 Seasonal Prediction Accuracy

From Table 2, M3 predicts best in **Jul** (RMSE = 3723.9) and worst in **Apr** (RMSE = 6409).

Summer months have the lowest errors because demand is consistently high and people tend to cycle regularly in good weather. The best-predicted month isn't necessarily the one with lowest demand, but the one where day-to-day variation is smallest. Mid-winter months also tend to be quite predictable since demand is low and steady.

Accuracy gets worse in spring (March and April) for a couple of reasons. First, spring demand depends on factors we don't model: Easter holidays, bank holidays, and sudden weather changes. Second, the post-COVID recovery was strongest in spring 2022, creating demand spikes that a smooth model can't handle. A planner using this model for March and April should add a buffer above predictions to account for this spring volatility.

5.2.3 Long-Term Trend and Extrapolation

Table 3: Trend coefficients from M1–M3.

Model	Daily change (cyclists/day)	Annual change (cyclists/year)
M1	-1.895	-692
M2	-1.872	-683

M3	-2.860	-1044

Annual growth rate. The M3 trend implies an annual change of **-1044 cyclists/day**, roughly **8.13%** of the predicted baseline count in January 2020. The trend is statistically significant ($p < 0.001$), but statistical significance doesn't mean we should take the number at face value.

Model consistency and COVID distortion. M1 and M2 yield roughly −690 cyclists/year, while M3 yields −1044. Including the COVID indicator separates the pandemic-period mean level from the secular trend, shifting the estimated slope. The pandemic years do not show notably low counts relative to 2023 and 2025, so there is no post-pandemic recovery to speak of; the trend is negative throughout the observation window. Re-estimation on post-COVID data only (2022–2025) yields a trend of **-2.857 cyclists/day** (**-1043 cyclists/year**), which is still **negative** ($p = 0$), confirming the decline is not simply a pandemic artefact. A piecewise or spline trend would give a more nuanced picture, but all specifications agree on the direction.

Extrapolation.

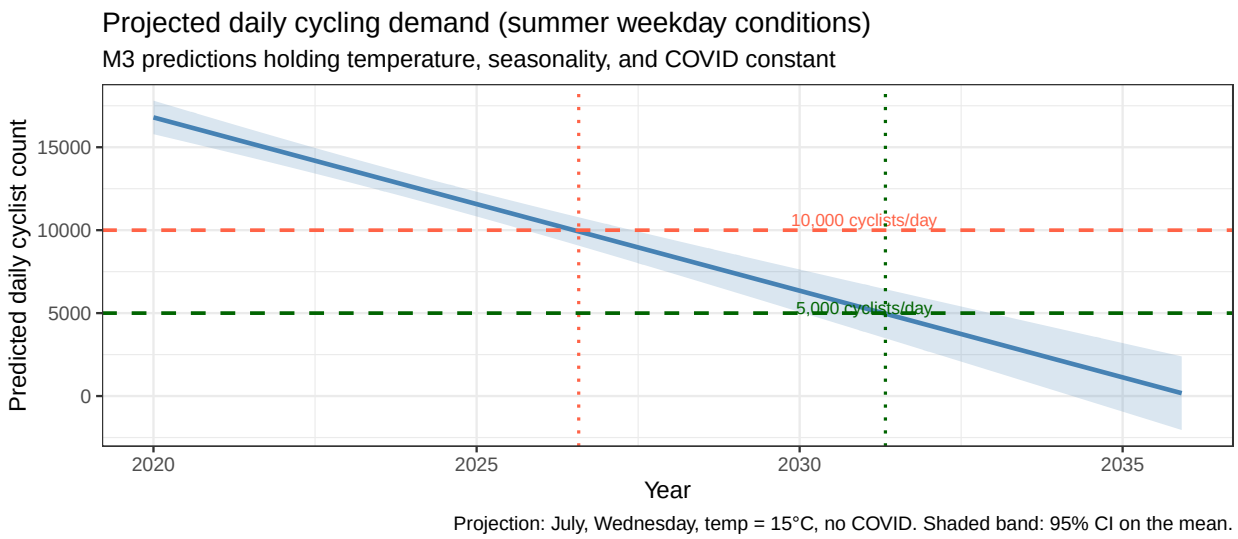

Figure 11: Projected summer weekday demand from M3 to 2035. Dashed lines mark planning thresholds at 10,000 and 5,000 cyclists/day. Shaded band: 95% CI on the mean.

Demand is projected to fall below 10,000 cyclists/day around **August 2026** and approach 5,000 by **May 2031**. This implausible result tells us the linear trend shouldn't be trusted for long-range forecasting. At least two reasons the extrapolation may not hold: (i) ongoing infrastructure investment could change cycling patterns in unpredictable ways; (ii) even if demand continues to decline, it would likely flatten out to a core group of committed cyclists rather than approach zero linearly. A spline or piecewise-linear trend would give very different projections.

5.2.4 Identifying a Poorly-Fit Period

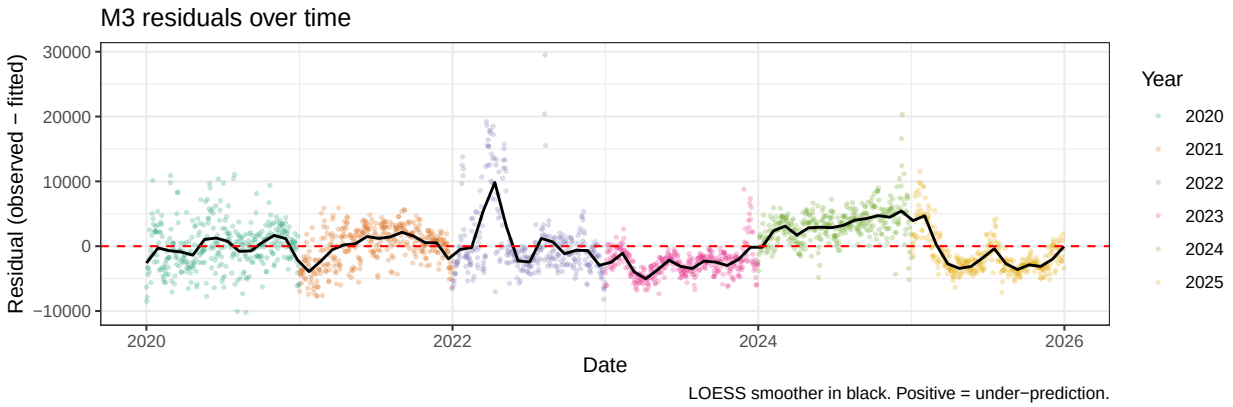

Figure 12: M3 residuals over time. The largest sustained under-prediction occurs in early-to-mid 2022.

Six month-year periods with largest absolute mean residual in M3.

Year	Month	Mean residual	Period RMSE
2022	Apr	7329	9993
2024	Dec	5483	7585
2022	Mar	5418	9601
2024	Oct	4846	5348
2023	Mar	-4676	4793
2023	Apr	-4630	4760

The worst single month is **Apr 2022**, with a mean residual of **7329 cyclists/day** and period RMSE of **9993 cyclists/day**, roughly 68% of the overall daily mean. A positive residual means M3 **under-predicts**: observed counts exceeded what the model expected. This makes sense for spring 2022, when post-COVID recovery picked up as people returned to offices but were still cautious about crowded public transport. The model's linear trend only captures the average rate of change over the full 2020–2025 window and can't track this faster recovery phase. A piecewise-linear trend with a changepoint around January 2022 would likely do a better job. The under-prediction reflects both a model limitation (not enough flexibility for rapid demand changes) and a data limitation (no event-level covariates like return-to-office mandates are available).

5.2.5 Original Diagnostic Plot

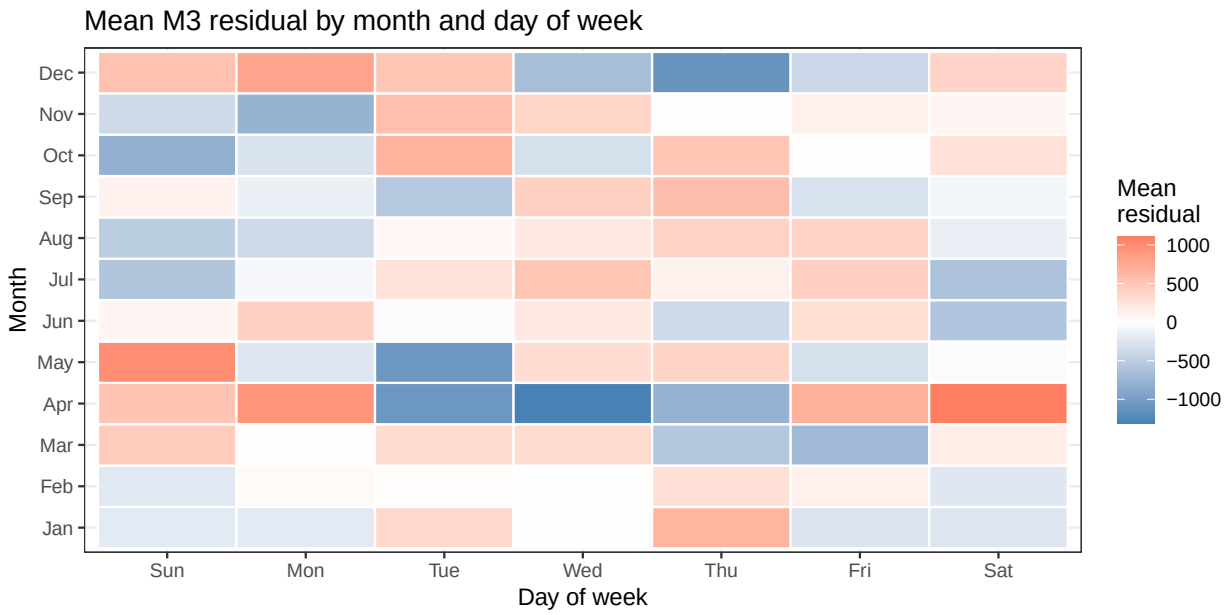

Figure 13: Mean M3 residual by month and day of week. Red = under-prediction (observed > fitted); blue = over-prediction (fitted > observed). The model under-predicts on Saturday, Sunday, and Monday in April and May while over-predicting on Tuesday and Wednesday in the same months, confirming that spring is where the additive model structure breaks down. A planner should conclude: (1) spring weekend and Monday capacity planning should add a buffer above predictions, and (2) a dow:season interaction term would improve spring predictions.

Figure 13 plots the mean M3 residual for every month and day-of-week cell. This isn't a standard diagnostic and doesn't appear elsewhere in the report. The most striking feature is the cluster of red cells (under-prediction) around Saturday, Sunday, and Monday in April and May. At the same time, Tuesday and Wednesday in April show blue cells, meaning the model over-predicts on those mid-week days. This isn't random noise; it reveals a structured interaction between day-of-week and season that the additive model can't capture. In spring, weekend and Monday demand is higher than the model expects, possibly because warmer weekends in April and May draw out large numbers of leisure cyclists that the model doesn't anticipate. The over-prediction on mid-week days in April could be linked to the April 2022 anomaly identified earlier, where the post-COVID surge was uneven across the week. A `dow:season` interaction term could help here. Planners relying on M3 for spring capacity should add a weekend buffer and treat mid-week spring predictions as slight over-estimates.

6 Discussion and Limitations

M3 explains 29.3% of daily count variance, which means over 70% of the day-to-day variation is unexplained. This is worth thinking about. The lag-1 autocorrelation of M3 residuals is 0.759, which is very high and tells us that consecutive days' errors are strongly correlated. In plain terms: if the model over-predicts today, it will probably over-predict tomorrow too. This happens because weather conditions persist over several days, and we're only using a single daily temperature as a covariate. Rainfall, wind speed, and daylight hours are all missing from the dataset but are known to affect cycling decisions. A time-series model with autoregressive errors, or a GAM with a smooth temporal component, would likely capture much of this unexplained variance. The R^2 is also deflated by extreme outlier days (festivals, unusual weather events) that

no model with these covariates would predict well. In terms of seasonal reliability, the model predicts best in Jul (RMSE = 3723.9) and worst in Apr (RMSE = 6409). Temperature sensitivity is about 228 additional cyclists per 1°C at 5°C, falling to roughly zero at 14.7°C. The estimated decline of 1044 cyclists/year should be treated with caution given the pandemic disruption in the data window; re-estimation on 2022–2025 data yields a negative trend of -1043 cyclists/year.

Model assumptions and the log-transform alternative. The Gaussian homoscedastic assumption isn't fully supported by the diagnostics. To check whether a variance-stabilising transformation resolves the Q-Q violation, we fit `lm(log(count) ~ ...)`:

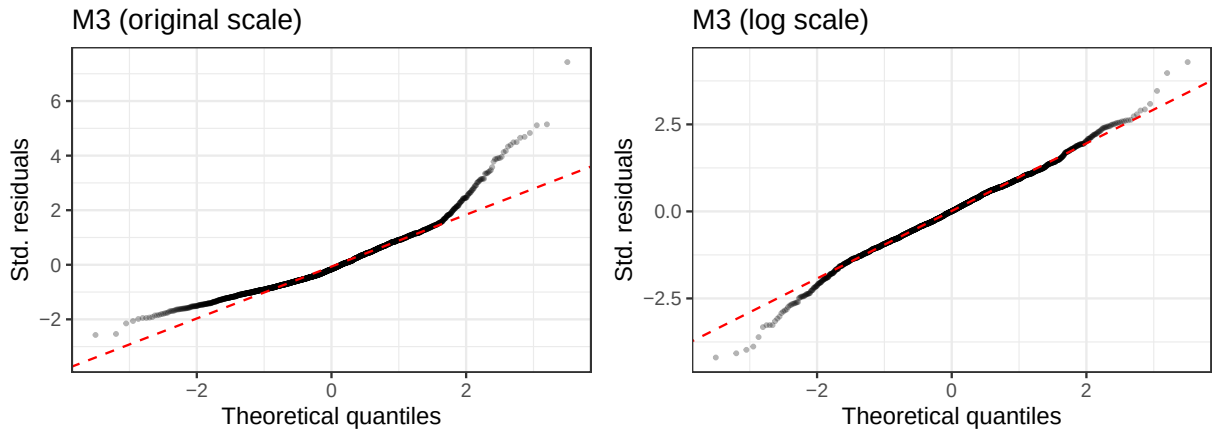

Figure 14: Q-Q comparison: M3 original (left) vs log-transformed (right). The log-transform removes the heavy right tail, confirming approximate log-normality of daily counts.

The log-transform does resolve the Q-Q departure ($R^2_{\log} = 0.356$ on the log scale). The problem is that when you back-transform predictions to the original count scale using $\hat{\mu} = \exp(\hat{\mu}_{\log} + \frac{1}{2}\hat{\sigma}^2_{\log})$, the prediction intervals get much wider, and IS penalises that. So we kept the untransformed response: the distributional fit isn't perfect, but the coefficients are directly readable in cyclists/day, which is what planners need. As a sanity check, a quasi-Poisson GLM gave a dispersion parameter of roughly 1500, confirming massive overdispersion. A negative binomial GLM would be a reasonable next step.

The empirical PI coverage (93.7% vs nominal 95%) is close to the target on average, but this hides the fact that intervals are too narrow for high-demand summer days and too wide for quiet winter days.

Key data limitations. Temperature is the only weather variable we have; rainfall, wind, and daylight hours aren't available. The lack of event-level covariates (bank holidays, school terms, festivals) partly explains why the model under-predicts on certain exceptional days. The COVID indicator is a fairly blunt tool that applies a constant mean shift across all of 2020–2021; in reality the impact varied month by month. Monthly COVID interactions or a time spline would capture the recovery pattern more faithfully. The residual heatmap (Figure 13) shows another limitation: the additive model misses the interaction between day-of-week and season, particularly in spring.

Generalisability. These results are specific to Edinburgh's aggregated counter network and its particular topography and climate. Changes in the number of active sensors over time could introduce measurement artefacts that look like demand trends. The most useful improvement would be replacing the linear trend with a piecewise-linear or spline trend that allows different growth rates before, during, and after the pandemic. Future work should also consider station-level modelling, sensor coverage adjustments, and additional weather variables.

7 Appendix: Code

All analysis code is in `code.R` and is reproduced below.

```
# ==============================================================================
# code.R -- Project 2: Modelling Daily Cycling Demand in Edinburgh
# Authors: Andreas Ioannou, Zacharias Ugochukwu Onyejiaka, Ioannis Alexandrides
# ==============================================================================
suppressPackageStartupMessages({
  library(tidyverse); library(knitr); library(kableExtra)
  library(patchwork); library(broom)
})
theme_set(theme_bw())

# ==============================================================================
# SECTION 1: Data Loading and Wrangling
# ==============================================================================

#' Load and prepare the cycling dataset
#' Converts month to ordered factor, re-specifies dow with Sunday as reference,
#' derives trend as integer days since 2020-01-01, and creates derived columns.
#' @param path character path to .Rdata file
#' @return data frame with all derived columns
load_cycling_data <- function(path = "cycle_daily_df.Rdata") {
  e <- new.env(); load(path, envir = e); df <- e$cycle_daily_df
  df %>% mutate(
    trend = as.integer(as.Date(date) - as.Date("2020-01-01")),
    month_num = as.integer(month), year = as.integer(year),
    month = factor(month, 1:12, c("Jan","Feb","Mar","Apr","May","Jun",
                                  "Jul","Aug","Sep","Oct","Nov","Dec"), ordered = TRUE),
    month_f = factor(as.character(month),
                     c("Jan","Feb","Mar","Apr","May","Jun",
                       "Jul","Aug","Sep","Oct","Nov","Dec"), ordered = FALSE),
    dow = factor(dow, c("Sun","Mon","Tue","Wed","Thu","Fri","Sat"), ordered = FALSE),
    temp_mean2 = temp_mean^2, temp_min2 = temp_min^2, temp_max2 = temp_max^2,
    is_covid = as.integer(year %in% 2020:2021),
    is_summer = as.integer(month_num %in% 6:8),
    season = factor(case_when(month_num %in% c(12,1,2) ~ "Winter",
                              month_num %in% 3:5 ~ "Spring",
                              month_num %in% 6:8 ~ "Summer",
                              month_num %in% 9:11 ~ "Autumn"),
                    c("Winter","Spring","Summer","Autumn")))
}

#' Summary statistics table for count and temperature variables
make_summary_table <- function(df) {
  vars <- list(count=df$count, temp_mean=df$temp_mean,
               temp_min=df$temp_min, temp_max=df$temp_max)
  do.call(rbind, lapply(names(vars), function(v) {
    x <- vars[[v]]
    data.frame(Variable=v, n=length(x), Mean=round(mean(x),2), SD=round(sd(x),2),
               Min=round(min(x),2), Q1=round(quantile(x,.25),2),
               Median=round(median(x),2), Q3=round(quantile(x,.75),2),
               Max=round(max(x),0), row.names=NULL)
```

```r
  }))
}

# ==============================================================================
# SECTION 2: EDA Plots
# ==============================================================================

#' Time series plot of daily counts with GAM smoother
plot_timeseries <- function(df) {
  ggplot(df, aes(as.Date(date), count)) +
    geom_line(colour="steelblue", alpha=.5, linewidth=.3) +
    geom_smooth(method="gam", formula=y~s(x), colour="red", se=FALSE, linewidth=.8) +
    labs(title="Daily cycling demand (Edinburgh, 2020--2025)",
         subtitle="Blue: daily counts; Red: GAM trend", x="Date", y="Daily cyclist count")
}

#' Boxplots of count by month, faceted by day of week
plot_month_dow_boxplot <- function(df) {
  ggplot(df, aes(x=month, y=count, fill=factor(weekend))) +
    geom_boxplot(alpha=.8, outlier.size=.3) + facet_wrap(~dow, ncol=4) +
    scale_fill_manual(values=c("0"="#6BAED6","1"="#E6550D"),
                      labels=c("Weekday","Weekend"), name="Day type") +
    labs(title="Cycling demand by month and day of week", x="Month", y="Daily cyclist count") +
    theme(axis.text.x=element_text(angle=60,hjust=1,size=7), legend.position="top")
}

#' Scatter plot of count vs mean temperature with quadratic smoother
plot_temp_scatter <- function(df) {
  ggplot(df, aes(temp_mean, count)) +
    geom_point(colour="grey60", alpha=.3, size=.7) +
    geom_smooth(method="lm", formula=y~x+I(x^2), colour="tomato", se=TRUE, linewidth=1) +
    labs(title="Cycling demand vs mean temperature",
         x="Mean temperature (\u00b0C)", y="Daily cyclist count")
}

#' Scatter plot coloured by meteorological season with linear fits
plot_season_scatter <- function(df) {
  ggplot(df, aes(temp_mean, count, colour=season)) +
    geom_point(alpha=.25, size=.8) +
    geom_smooth(method="lm", formula=y~x, se=FALSE, linewidth=1.2) +
    scale_colour_manual(values=c(Winter="#2171B5",Spring="#41AB5D",
                                 Summer="#D7301F",Autumn="#FD8D3C"), name="Season") +
    labs(title="Temperature sensitivity by season",
         x="Mean temperature (\u00b0C)", y="Daily cyclist count")
}

#' Heatmap of mean daily count by year and month
plot_count_heatmap <- function(df) {
  df %>% group_by(year,month) %>% summarise(mean_count=mean(count),.groups="drop") %>%
    ggplot(aes(factor(year), month, fill=mean_count)) +
    geom_tile(colour="white",linewidth=.4) +
    scale_fill_viridis_c(option="C", name="Mean\ncount") +
    labs(title="Mean daily cyclist count by year and month", x="Year", y="Month")
```

```r
}

# ==============================================================================
# SECTION 3: Model Fitting
# ==============================================================================

#' M0: baseline with numeric month and binary weekend
fit_M0 <- function(df) lm(count ~ temp_mean + weekend + month_num, data=df)

#' M1: factor month, factor dow, linear trend
fit_M1 <- function(df) lm(count ~ temp_mean + month_f + trend + dow, data=df)

#' M2: adds quadratic temperature to M1
fit_M2 <- function(df) lm(count ~ temp_mean + temp_mean2 + month_f + trend + dow, data=df)

#' M3: adds COVID indicator to M2
fit_M3 <- function(df) lm(count ~ temp_mean + temp_mean2 + month_f + trend + dow + is_covid, data=df)

#' M3 on log scale (for Discussion comparison)
fit_M3_log <- function(df) lm(log(count) ~ temp_mean + temp_mean2 + month_f + trend + dow + is_covid, da

# ==============================================================================
# SECTION 4: Diagnostic Plots
# ==============================================================================

#' Four-panel diagnostic: residuals-vs-fitted, QQ, scale-location,
#' residuals over time coloured by year
diagnostic_plots <- function(fit, df, title="") {
  dd <- data.frame(fitted=fitted(fit), residual=residuals(fit),
                   std_res=rstandard(fit), date=as.Date(df$date), year=factor(df$year))
  p1 <- ggplot(dd,aes(fitted,residual))+geom_point(alpha=.2,size=.6)+
    geom_hline(yintercept=0,colour="red",linetype="dashed")+
    geom_smooth(method="loess",formula=y~x,se=FALSE,colour="steelblue",linewidth=.8)+
    labs(title="Residuals vs Fitted",x="Fitted values",y="Residuals")
  p2 <- ggplot(dd,aes(sample=std_res))+stat_qq(alpha=.3,size=.6)+
    stat_qq_line(colour="red",linetype="dashed")+
    labs(title="Normal Q-Q",x="Theoretical quantiles",y="Standardised residuals")
  p3 <- ggplot(dd,aes(fitted,sqrt(abs(std_res))))+geom_point(alpha=.2,size=.6)+
    geom_smooth(method="loess",formula=y~x,se=FALSE,colour="steelblue",linewidth=.8)+
    labs(title="Scale-Location",x="Fitted values",y=expression(sqrt("|Std. residuals|")))
  p4 <- ggplot(dd,aes(date,residual,colour=year))+
    geom_line(linewidth=.3,alpha=.7)+
    geom_hline(yintercept=0,colour="red",linetype="dashed")+
    scale_colour_brewer(palette="Dark2",name="Year")+
    labs(title="Residuals over time",x="Date",y="Residuals")+theme(legend.position="right")
  (p1|p2)/(p3|p4)+plot_annotation(title=paste(title,"diagnostics"),tag_levels="a")&theme_bw(base_size=1
}

#' Coefficient table from a fitted lm object
coef_table <- function(fit, digits=3) {
  s <- summary(fit)$coefficients
  data.frame(Term=rownames(s), Estimate=round(s[,"Estimate"],digits),
             Std.Error=round(s[,"Std. Error"],digits),
```

```r
             t.value=round(s[,"t value"],2),
             p.value=signif(s[,"Pr(>|t|)"],3), row.names=NULL)
}

# ==============================================================================
# SECTION 5: Scoring & Cross-Validation
# ==============================================================================

#' Compute RMSE, MAE, DS and IS from predictive summaries
calc_scores <- function(y, mu, sig, alpha=.05) {
  e <- y-mu; z <- qnorm(1-alpha/2); lo <- mu-z*sig; hi <- mu+z*sig
  is_val <- (hi-lo)+(2/alpha)*pmax(lo-y,0)+(2/alpha)*pmax(y-hi,0)
  c(RMSE=sqrt(mean(e^2)),MAE=mean(abs(e)),DS=mean(log(sig^2)+e^2/sig^2),IS=mean(is_val))
}

#' Leave-one-year-out CV for all four models
#' Uses month_num (numeric) to avoid missing-factor-level errors
run_loyo_cv_all <- function(df) {
  cv_forms <- list(M0=count~temp_mean+weekend+month_num,
                   M1=count~temp_mean+month_num+trend+dow,
                   M2=count~temp_mean+temp_mean2+month_num+trend+dow,
                   M3=count~temp_mean+temp_mean2+month_num+trend+dow+is_covid)
  years <- sort(unique(df$year))
  do.call(rbind, lapply(names(cv_forms), function(mn) {
    fs <- do.call(rbind, lapply(years, function(yr) {
      tr <- df[df$year!=yr,,drop=FALSE]; te <- df[df$year==yr,,drop=FALSE]
      fit <- lm(cv_forms[[mn]],data=tr); pr <- predict(fit,newdata=te,se.fit=TRUE)
      sig <- sqrt(summary(fit)$sigma^2+pr$se.fit^2)
      calc_scores(te$count,pr$fit,sig)
    }))
    data.frame(Model=mn,RMSE=round(mean(fs[,"RMSE"]),1),MAE=round(mean(fs[,"MAE"]),1),
               DS=round(mean(fs[,"DS"]),2),IS=round(mean(fs[,"IS"]),1),row.names=NULL)
  }))
}

#' Empirical PI coverage for each model
compute_pi_coverage <- function(df) {
  cv_forms <- list(M0=count~temp_mean+weekend+month_num,
                   M1=count~temp_mean+month_num+trend+dow,
                   M2=count~temp_mean+temp_mean2+month_num+trend+dow,
                   M3=count~temp_mean+temp_mean2+month_num+trend+dow+is_covid)
  years <- sort(unique(df$year))
  sapply(names(cv_forms), function(mn) {
    cv <- sapply(years, function(yr) {
      tr <- df[df$year!=yr,,drop=FALSE]; te <- df[df$year==yr,,drop=FALSE]
      fit <- lm(cv_forms[[mn]],data=tr); pr <- predict(fit,newdata=te,se.fit=TRUE)
      sig <- sqrt(summary(fit)$sigma^2+pr$se.fit^2); z <- qnorm(.975)
      mean(te$count>=(pr$fit-z*sig) & te$count<=(pr$fit+z*sig))
    }); round(mean(cv)*100,1)
  })
}

#' Monthly CV scores for the final model (M3)
```

```r
monthly_cv_scores <- function(df) {
  f <- count~temp_mean+temp_mean2+month_num+trend+dow+is_covid
  years <- sort(unique(df$year))
  ml <- c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec")
  ao <- do.call(rbind, lapply(years, function(yr) {
    tr <- df[df$year!=yr,,drop=FALSE]; te <- df[df$year==yr,,drop=FALSE]
    fit <- lm(f,data=tr); pr <- predict(fit,newdata=te,se.fit=TRUE)
    sig <- sqrt(summary(fit)$sigma^2+pr$se.fit^2)
    data.frame(month_num=te$month_num,y=te$count,mu=pr$fit,sig=sig)
  }))
  ao %>% group_by(month_num) %>%
    summarise(RMSE=round(sqrt(mean((y-mu)^2)),1),
              DS=round(mean(log(sig^2)+(y-mu)^2/sig^2),2),.groups="drop") %>%
    arrange(month_num) %>% mutate(Month=ml[month_num]) %>% select(Month,RMSE,DS)
}

#' CV comparison of M3 with temp_mean, temp_min, temp_max
compare_temp_variants <- function(df) {
  years <- sort(unique(df$year))
  do.call(rbind, lapply(c("temp_mean","temp_min","temp_max"), function(tv) {
    tv2 <- paste0(tv,"2")
    f <- as.formula(paste("count~",tv,"+",tv2,"+ month_num+trend+dow+is_covid"))
    sc <- do.call(rbind, lapply(years, function(yr) {
      tr <- df[df$year!=yr,,drop=FALSE]; te <- df[df$year==yr,,drop=FALSE]
      fit <- lm(f,data=tr); pr <- predict(fit,newdata=te,se.fit=TRUE)
      sig <- sqrt(summary(fit)$sigma^2+pr$se.fit^2)
      calc_scores(te$count,pr$fit,sig)
    }))
    data.frame(`Temperature variable`=tv, RMSE=round(mean(sc[,"RMSE"]),1),
               DS=round(mean(sc[,"DS"]),2), IS=round(mean(sc[,"IS"]),1),
               check.names=FALSE, row.names=NULL)
  }))
}

# ==============================================================================
# SECTION 6: Marginal Effect & Planning
# ==============================================================================

#' Marginal effect of temperature at a given point
marginal_effect <- function(fit, temp_lin, temp_quad, T0) {
  cf <- coef(fit); as.numeric(cf[temp_lin]+2*cf[temp_quad]*T0)
}

#' Plot marginal effect curve with annotations at 5C and 15C
plot_marginal_temp <- function(fit, df, tv="temp_mean") {
  cf <- coef(fit); tv2 <- paste0(tv,"2"); b1 <- cf[tv]; b2 <- cf[tv2]
  T_star <- round(-b1/(2*b2),1)
  e5 <- round(as.numeric(b1+2*b2*5),0); e15 <- round(as.numeric(b1+2*b2*15),0)
  ts <- seq(floor(min(df[[tv]]))-1, ceiling(max(df[[tv]]))+1, by=.2)
  mdf <- data.frame(temp=ts, mfx=as.numeric(b1+2*b2*ts))
  ggplot(mdf,aes(temp,mfx))+geom_line(colour="steelblue",linewidth=1.2)+
    geom_hline(yintercept=0,linetype="dashed",colour="grey40",linewidth=.8)+
    geom_vline(xintercept=c(5,15),linetype="dotted",colour="tomato",linewidth=.8)+
```

```r
    geom_point(data=data.frame(x=c(5,15),y=as.numeric(b1+2*b2*c(5,15))),
               aes(x,y),colour="tomato",size=3.5)+
    annotate("text",x=5,y=as.numeric(b1+2*b2*5)+20,label=paste0(e5," cyc/\u00b0C"),
             colour="tomato",size=3.2,vjust=0)+
    annotate("text",x=15,y=as.numeric(b1+2*b2*15)+20,label=paste0(e15," cyc/\u00b0C"),
             colour="tomato",size=3.2,vjust=0)+
    annotate("text",x=as.numeric(T_star),y=8,label=paste0("T* = ",T_star,"\u00b0C"),
             colour="grey30",size=3,hjust=-.1)+
    labs(title=paste0("Marginal effect of ",tv," on daily cyclist count (M3)"),
         x=paste0(tv," (\u00b0C)"), y="Marginal effect (cyclists per 1\u00b0C increase)")
}

#' Table of trend coefficients for M1-M3
trend_table <- function(m1,m2,m3) {
  dv <- c(M1=as.numeric(coef(m1)["trend"]),M2=as.numeric(coef(m2)["trend"]),
          M3=as.numeric(coef(m3)["trend"]))
  do.call(rbind, lapply(names(dv), function(nm)
    data.frame(Model=nm, `Daily change (cyclists/day)`=round(dv[nm],3),
               `Annual change (cyclists/year)`=round(dv[nm]*365,0),
               check.names=FALSE, row.names=NULL)))
}

#' Re-estimate trend on post-COVID data only (2022-2025)
post_covid_trend <- function(df) {
  post_df <- df %>% filter(year >= 2022)
  fit <- lm(count ~ temp_mean+temp_mean2+month_f+trend+dow, data=post_df)
  s <- summary(fit)$coefficients
  list(daily=round(coef(fit)["trend"],3), annual=round(coef(fit)["trend"]*365,0),
       pval=round(s["trend","Pr(>|t|)"],4),
       dir=ifelse(coef(fit)["trend"]>0,"positive","negative"))
}

#' Project demand forward under fixed conditions
project_demand <- function(fit, df, end_year=2035) {
  dates <- seq(as.Date("2020-01-01"),as.Date(paste0(end_year,"-12-31")),by="month")
  nd <- data.frame(trend=as.integer(dates-as.Date("2020-01-01")),
                   temp_mean=15,temp_mean2=225,weekend=0L,
                   month_f=factor("Jul",levels=levels(df$month_f)),
                   dow=factor("Wed",levels=levels(df$dow)),month_num=7L,is_covid=0L)
  pred <- predict(fit,newdata=nd,interval="confidence")
  data.frame(date=dates,fit=pred[,"fit"],lwr=pred[,"lwr"],upr=pred[,"upr"])
}

#' Find date at which projected demand crosses a threshold
threshold_date <- function(proj_df,threshold,direction="below") {
  cross <- if(direction=="below") proj_df[proj_df$fit<=threshold,]
  else proj_df[proj_df$fit>=threshold,]
  if(nrow(cross)>0) format(cross$date[1],"%B %Y") else "beyond projection window"
}

#' Plot extrapolation with threshold lines
plot_extrapolation <- function(proj_df, thresholds=c(10000,5000)) {
  cols <- c("tomato","darkgreen")
```

```
  p <- ggplot(proj_df,aes(date))+
    geom_ribbon(aes(ymin=lwr,ymax=upr),fill="steelblue",alpha=.2)+
    geom_line(aes(y=fit),colour="steelblue",linewidth=1)
  for(i in seq_along(thresholds)){
    th <- thresholds[i]
    p <- p+geom_hline(yintercept=th,linetype="dashed",colour=cols[i],linewidth=.8)+
      annotate("text",x=as.Date("2031-01-01"),y=th*1.06,
               label=paste0(format(th,big.mark=","),"  cyclists/day"),colour=cols[i],size=2.8)
    cross <- proj_df[proj_df$fit<=th,]
    if(nrow(cross)>0) p <- p+geom_vline(xintercept=as.numeric(cross$date[1]),
                                        linetype="dotted",colour=cols[i],linewidth=.7)
  }
  p+labs(title="Projected daily cycling demand (summer weekday conditions)",
         subtitle="M3 predictions holding temperature, seasonality, and COVID constant",
         x="Year",y="Predicted daily cyclist count",
         caption="Projection: July, Wednesday, temp = 15\u00b0C, no COVID. Shaded band: 95% CI on the m
}

# ==============================================================================
# SECTION 7: Poorly-Fit Period & Original Diagnostic
# ==============================================================================

#' Identify worst-predicted month-year periods
worst_periods <- function(fit, df, n=6) {
  labs <- c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec")
  data.frame(residual=residuals(fit),year=df$year,month_num=df$month_num) %>%
    group_by(year,month_num) %>%
    summarise(`Mean residual`=round(mean(residual),0),
              `Period RMSE`=round(sqrt(mean(residual^2)),0),.groups="drop") %>%
    arrange(desc(abs(`Mean residual`))) %>% head(n) %>%
    mutate(Month=labs[month_num]) %>% select(Year=year,Month,`Mean residual`,`Period RMSE`)
}

#' Plot residuals over time with LOESS smoother
plot_resid_time <- function(fit, df) {
  data.frame(date=as.Date(df$date),residual=residuals(fit),year=factor(df$year)) %>%
    ggplot(aes(date,residual,colour=year))+geom_point(alpha=.25,size=.6)+
    geom_smooth(aes(group=1),method="loess",span=.05,formula=y~x,
                colour="black",se=FALSE,linewidth=.7)+
    geom_hline(yintercept=0,linetype="dashed",colour="red")+
    scale_colour_brewer(palette="Dark2",name="Year")+
    labs(title="M3 residuals over time",x="Date",y="Residual (observed \u2212 fitted)",
         caption="LOESS smoother in black. Positive = under-prediction.")
}

#' Heatmap of mean residuals by month x day-of-week
plot_resid_heatmap <- function(fit, df) {
  data.frame(residual=residuals(fit),month=df$month,dow=df$dow) %>%
    group_by(month,dow) %>% summarise(mean_resid=mean(residual),.groups="drop") %>%
    ggplot(aes(x=dow,y=month,fill=mean_resid))+
    geom_tile(colour="white",linewidth=.5)+
    scale_fill_gradient2(low="steelblue",mid="white",high="tomato",
                         midpoint=0,name="Mean\nresidual")+
```

```r
    labs(title="Mean M3 residual by month and day of week",x="Day of week",y="Month")
}

#' QQ comparison between M3 on original scale and log scale
plot_qq_comparison <- function(m3, m3_log) {
  p1 <- ggplot(data.frame(s=rstandard(m3)),aes(sample=s))+stat_qq(alpha=.3,size=.6)+
    stat_qq_line(colour="red",linetype="dashed")+
    labs(title="M3 (original scale)",x="Theoretical quantiles",y="Std. residuals")
  p2 <- ggplot(data.frame(s=rstandard(m3_log)),aes(sample=s))+stat_qq(alpha=.3,size=.6)+
    stat_qq_line(colour="red",linetype="dashed")+
    labs(title="M3 (log scale)",x="Theoretical quantiles",y="Std. residuals")
  p1|p2
}
```